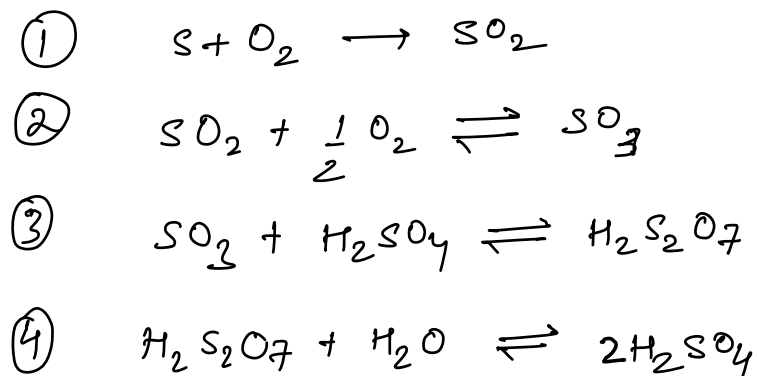


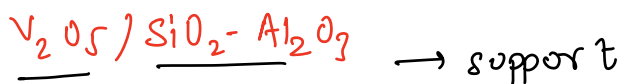
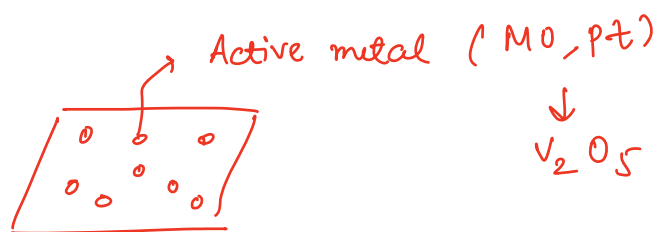
Course Content

- Some basic chemical reactions relevant to process industries.
- S and H_2SO_4 manufacturing .(CONTACT PROCESS)
- polymer fundamentals.
- Petroleum refinery industries.
- catalyst preparation (supported catalyst) (*)

CONTACT PROCESS (H_2SO_4 Manufacturing):



(*)



↓
Metal
oxide

Sulphur Manufacturing:

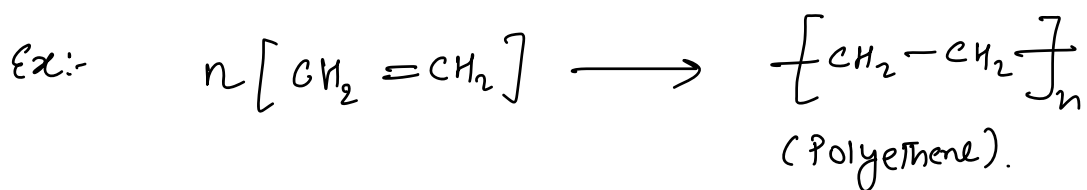
- ① FRASCH Process
- ② H_2S (Most important).
- ③ FINNISH Process (uses Iron Pyrites (FeS_2)).

Polymer Fundamentals:

Monomeric unit should have

- ① = (functionality = 2)
- ② $\equiv$ (functionality = 4).
- ③ Reactive Functional Groups
 - $COOH$
 - NH_2
 - OH

Dimer
↓
Trimer
↓
Tetramer
⋮
⋮

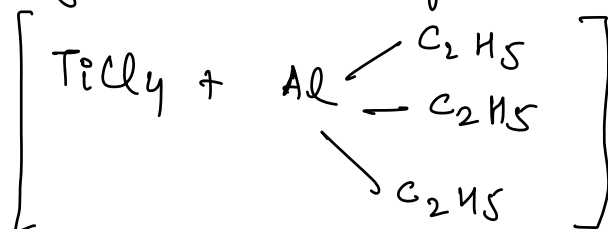


3 processes of producing polyethylene:

- ① High pressure process
- ② Intermediate pressure process
- ③ Low pressure process

→
a) $70^\circ C$
b) 6-10 atm
c) catalyst

Ziegler - Natta catalyst



↓
Ziegler - Natta catalyst

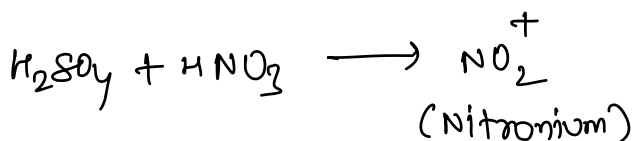
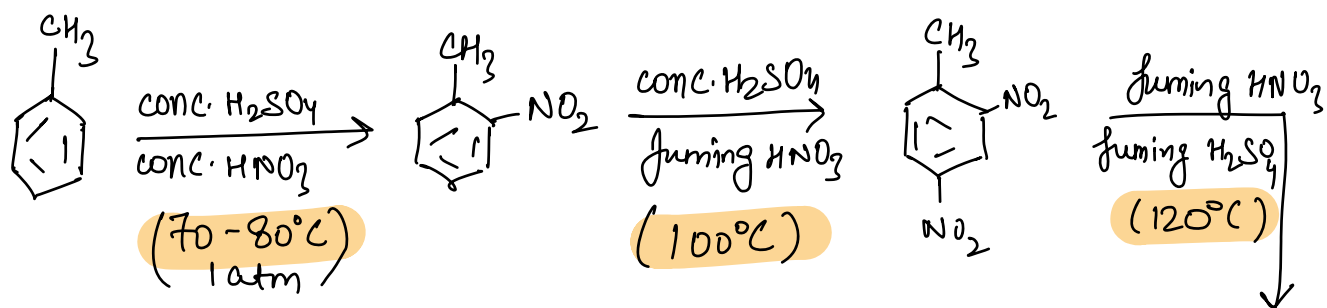
Petroleum Refinery Industries:

① Saturated compounds (paraffin) (Ex:- n-Heptane)

② Unsaturated compounds ($=$, $\equiv$)

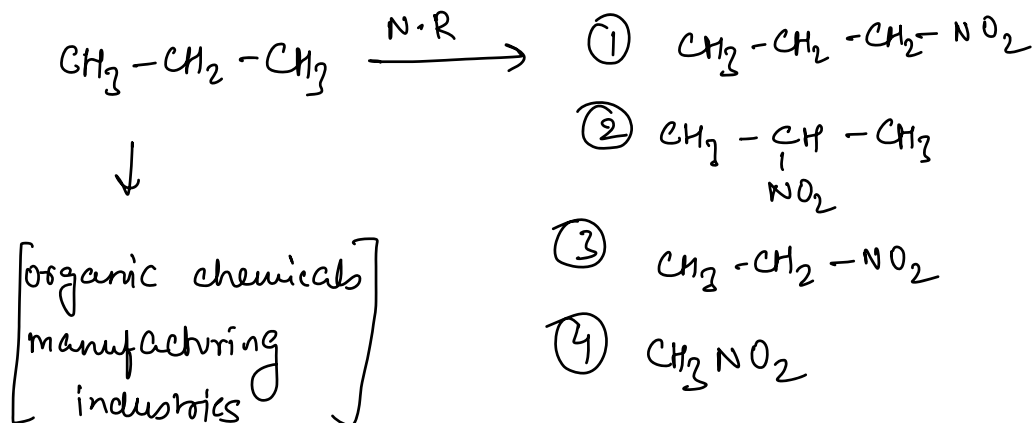
③ Naphthenic compounds (C_nH_{2n}) 

④ Aromatic compounds 



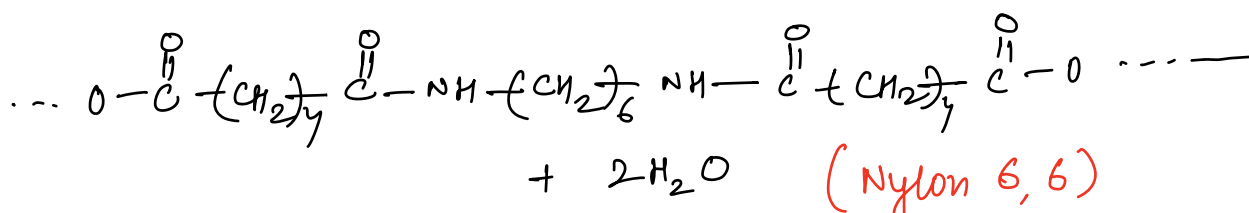
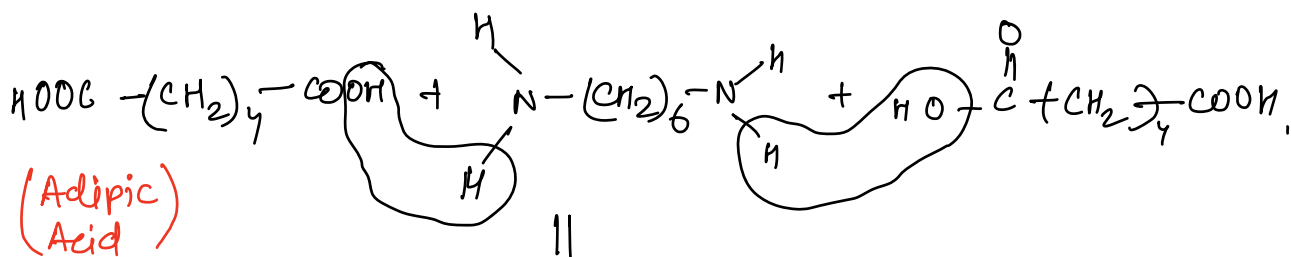
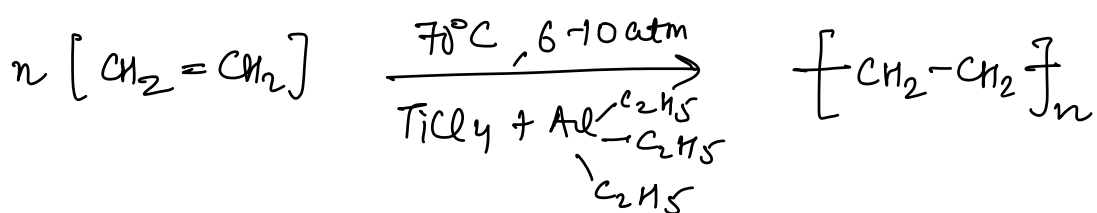
2,4,6-trinitrotoluene

Nitration Rxn



Polymerization Rxn

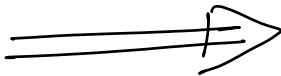
- | | | |
|--|---|--|
| ① Addition polymerization
② Condensation polymerization |] | By products (X)
By products (✓)
($\text{CO}_2, \text{NH}_3, \text{H}_2\text{O} \dots$) |
|--|---|--|



H.W

Claus process

(To produce Sulphur)
from H_2S



Claus process

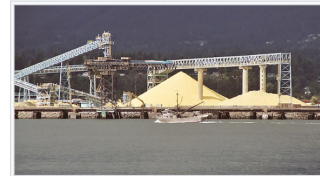
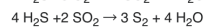
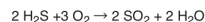
From Wikipedia, the free encyclopedia

The **Claus process** is the most significant gas **desulfurizing** process, recovering elemental **sulfur** from gaseous **hydrogen sulfide**. First patented in 1883 by the chemist **Carl Friedrich Claus**, the Claus process has become the industry standard.

The multi-step Claus process recovers sulfur from the gaseous hydrogen sulfide found in raw **natural gas** and from the by-product gases containing hydrogen sulfide derived from refining **crude oil** and other industrial processes. The by-product gases mainly originate from physical and chemical gas treatment units (**Selexol**, **Rectisol**, **Purisol** and **amine scrubbers**) in refineries, **natural gas processing plants** and **gasification** or **synthesis gas plants**. These by-product gases may also contain **hydrogen cyanide**, **hydrocarbons**, **sulfur dioxide** or **ammonia**.

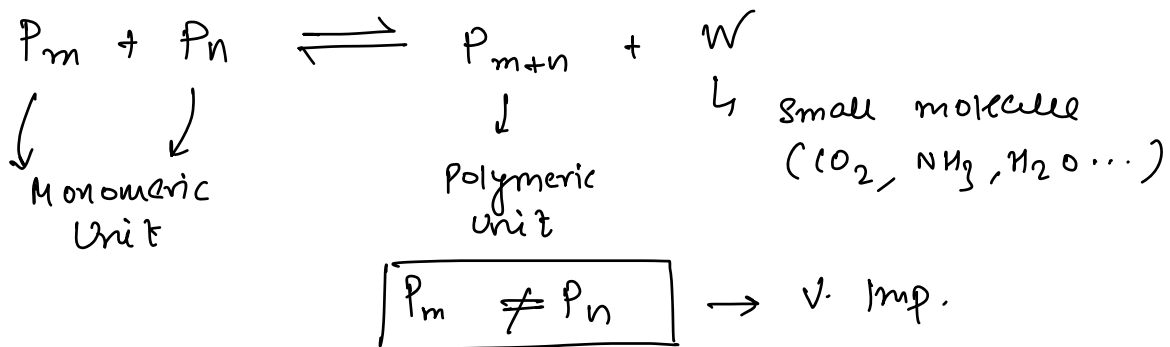
Gases with an H_2S content of over 25% are suitable for the recovery of sulfur in straight-through Claus plants while alternate configurations such as a split-flow set up or feed and air preheating can be used to process leaner feeds.^[1]

Hydrogen sulfide produced, for example, in the **hydro-desulfurization** of refinery **naphthas** and other **petroleum** oils, is converted to sulfur in Claus plants.^[2] The reaction proceeds in two steps:

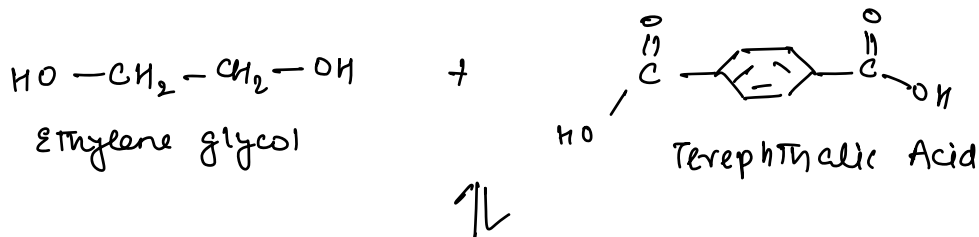


Piles of sulfur produced in Alberta by the Claus process awaiting shipment at docks in Vancouver, Canada.

Condensation polymerization continued



(#)



PET
(Poly ethylene terephthalate) $\rightarrow$ Alkyl Resin

$\downarrow$
RESIN MANUFACTURING
INDUSTRIES.

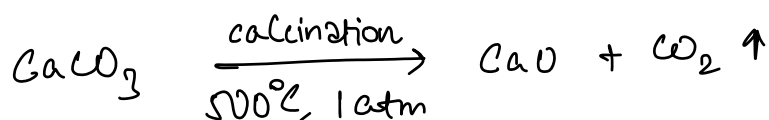
CALCINATION

- Used in cement manufacturing industries.

- calcination $\left\{ \begin{array}{l} \rightarrow \text{Absence of } \text{Air} \text{ (Inert environment - } \text{N}_2, \text{Ar, He}) \\ \rightarrow \text{Presence of } \text{Air} \text{ (O}_2 \text{ / Air environment)} \end{array} \right.$

$\downarrow$
cheap

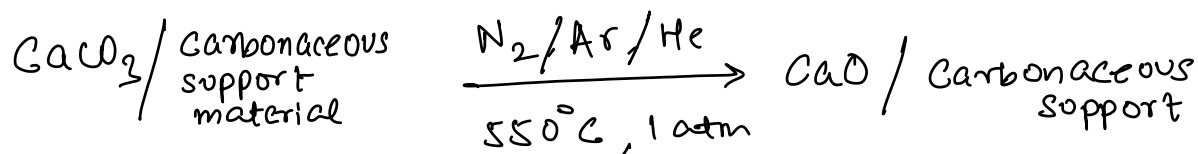
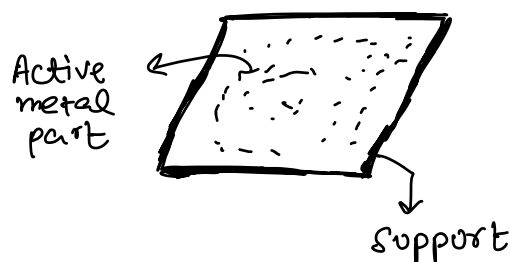
In the presence of Air $\Rightarrow$

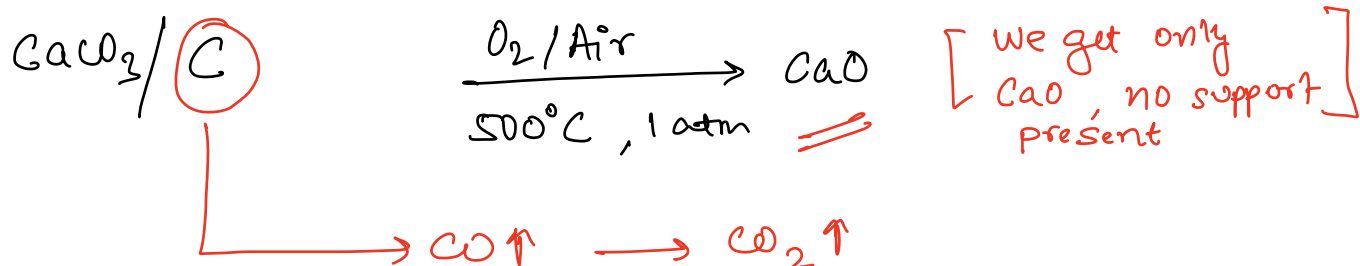


In the absence of Air $\Rightarrow$

① ACTIVE METAL OXIDE

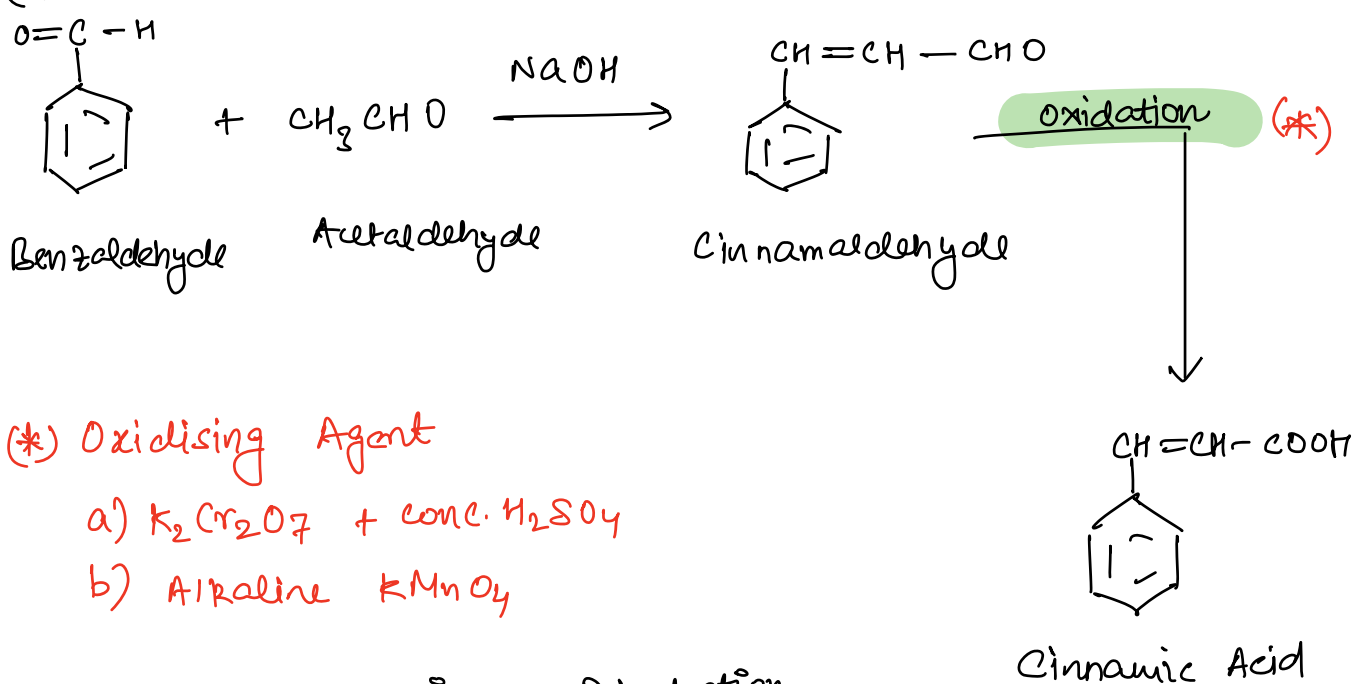
② Support (Medium to High Surface Area)
(BET method, m^2/g)



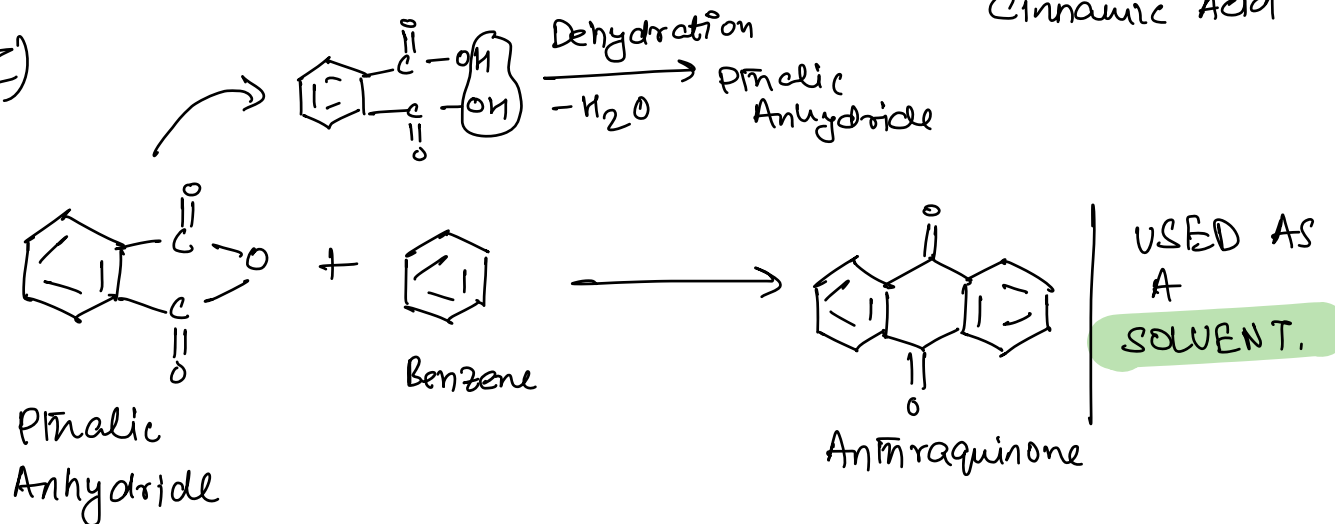


CONDENSATION [organic chemical manufacturing industries]

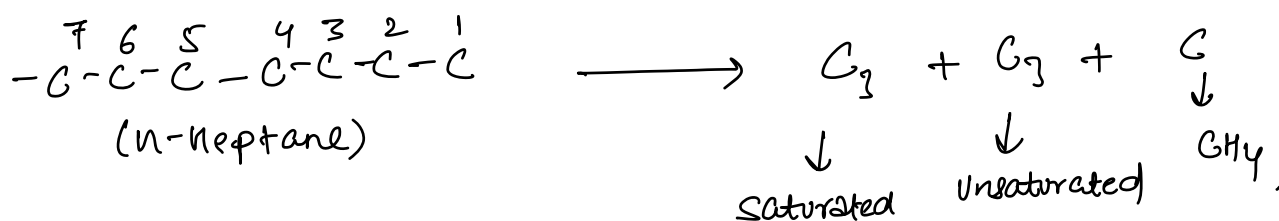
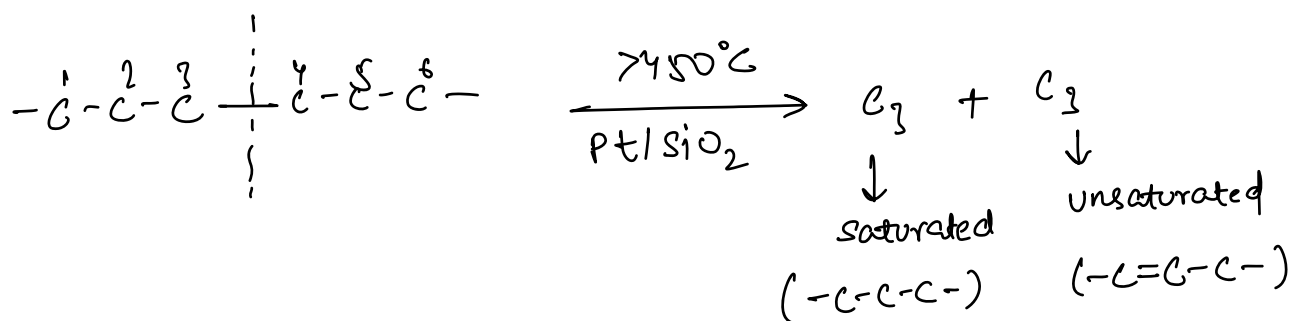
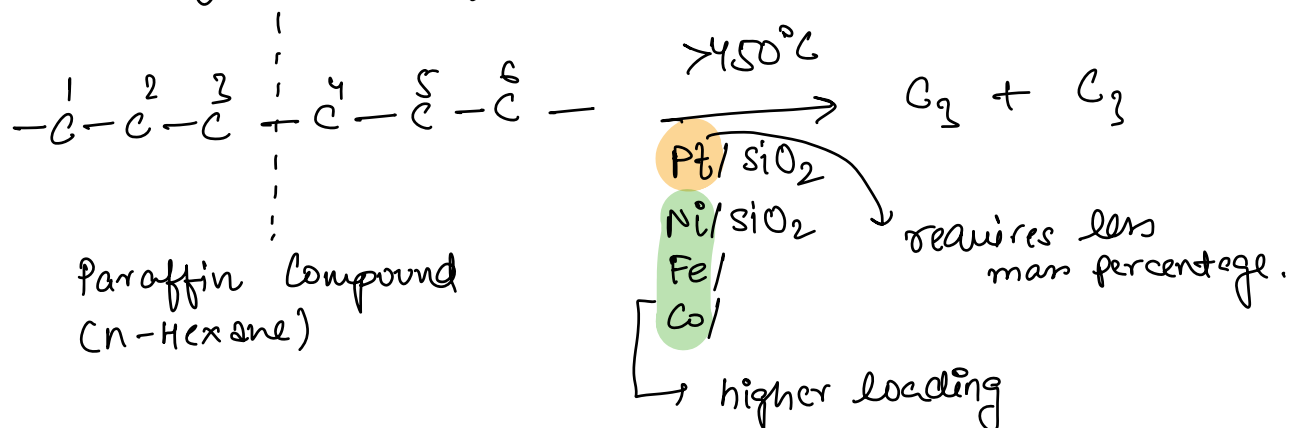
(I)



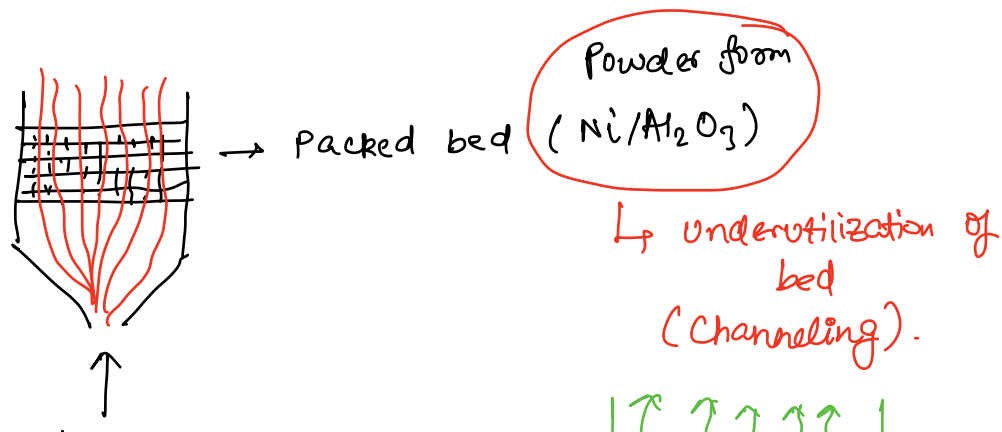
(II)



(III) Catalytic Cracking (Used in petroleum Industries)

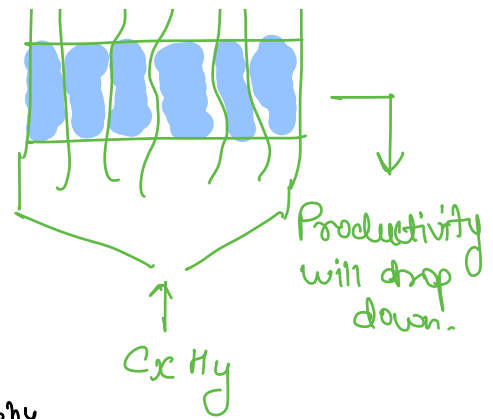


(#)

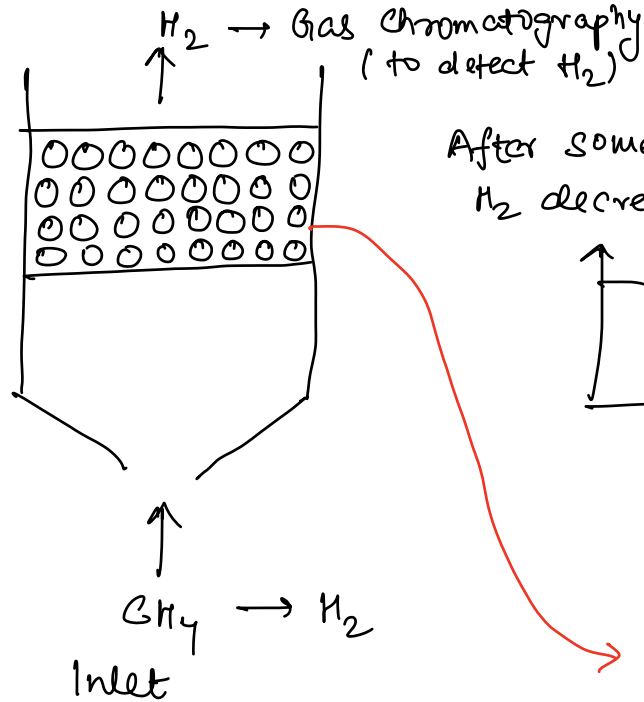


Inlet
 C_xH_y
 (Hydrocarbon)

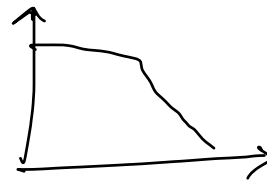
→ whereas Pellets (3mm) are more uniformly distributed.



NOW



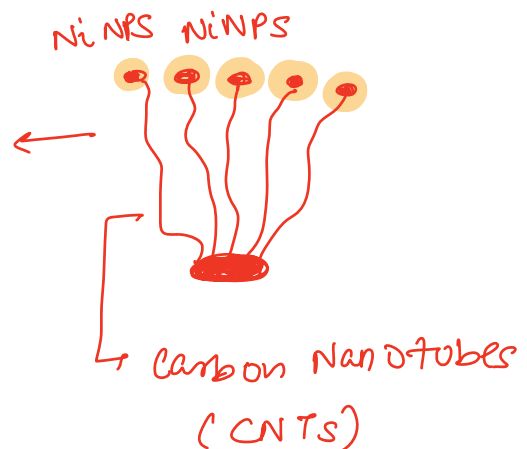
After some time production of H_2 decreases



→ If observed under electron microscope

Ni NPS → nickel nanoparticles

Amorphous in nature



→ Tip Growth (Metal deposits on Tip)
 (less interaction btw active metal and support particles)

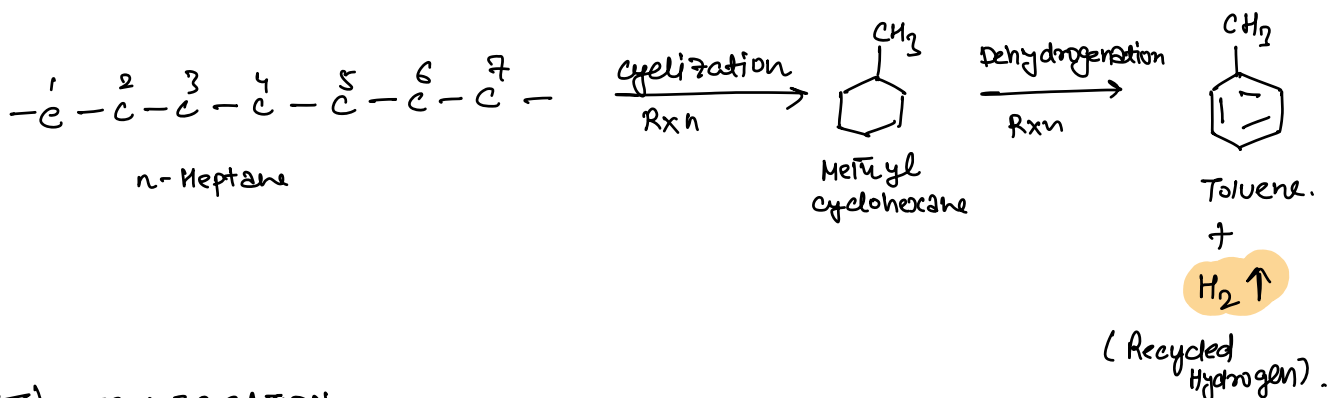
CNT PRODUCTION

Base Growth (Metal deposits on base).
(more interaction btw active metal and support particle)

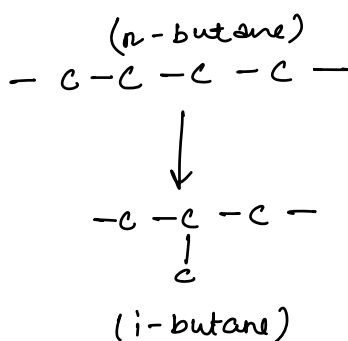
Surface area decreases because of formation of Amorphous Carbon Layer, thus decreasing productivity.

(IV) CATALYTIC REFORMING (Petroleum Refineries)

Paraffin dehydrocyclization $\begin{cases} \rightarrow \text{Dehydrogenation} \\ \rightarrow \text{Cyclization} \end{cases}$



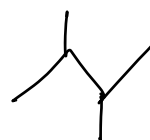
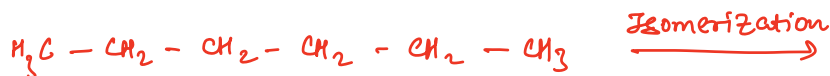
(V) ISOMERISATION



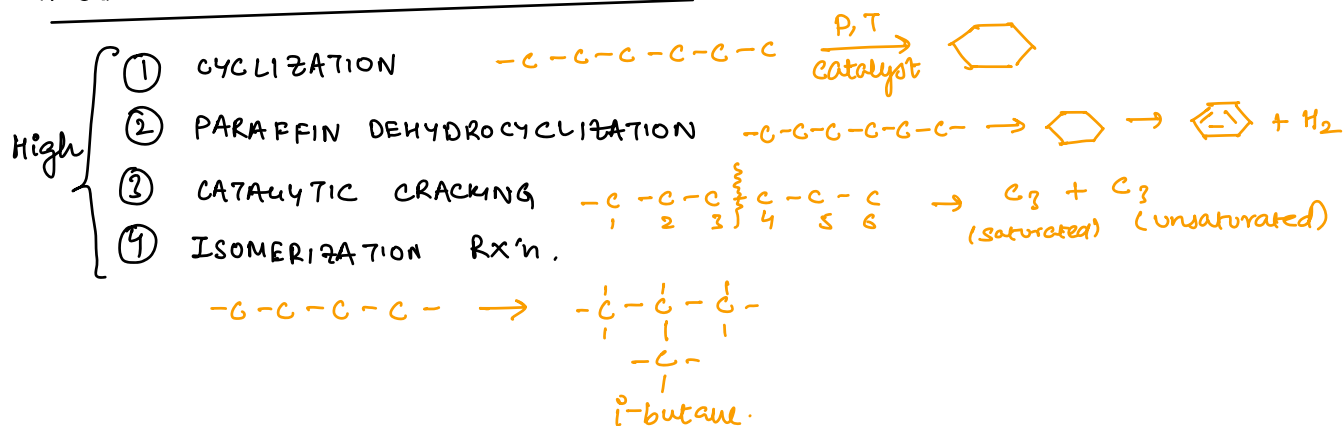
Total no. of C and H atoms remain same



HW

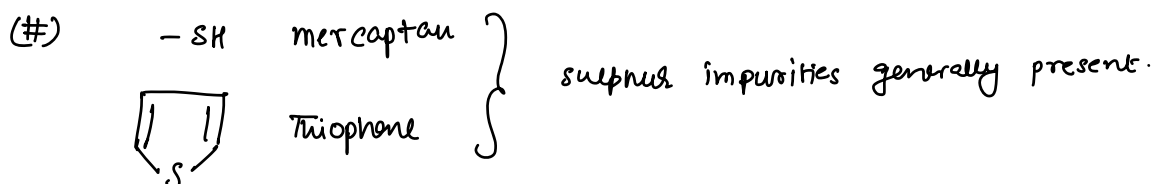


Reactions Relevant to Petroleum Refineries:-



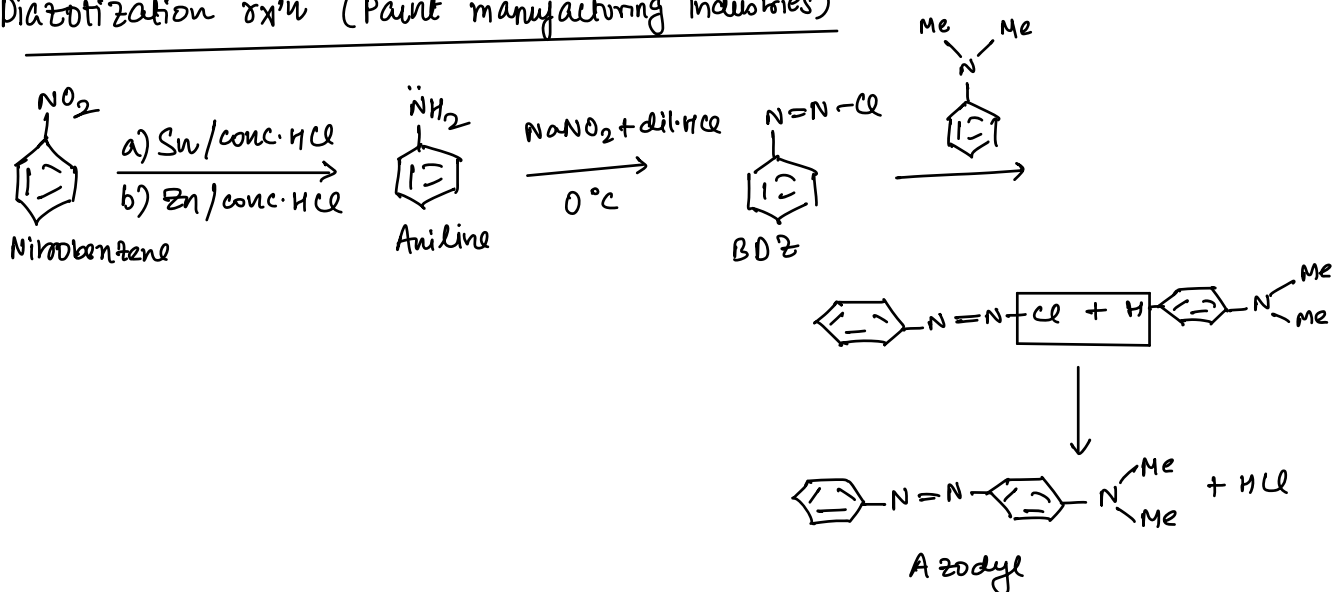
Low $\rightarrow$ when feed contains (S, O, N) impurities.

Feed (S, O, N) $\rightarrow$ Purification $\rightarrow$ Purified feed stream $\rightarrow$ PR

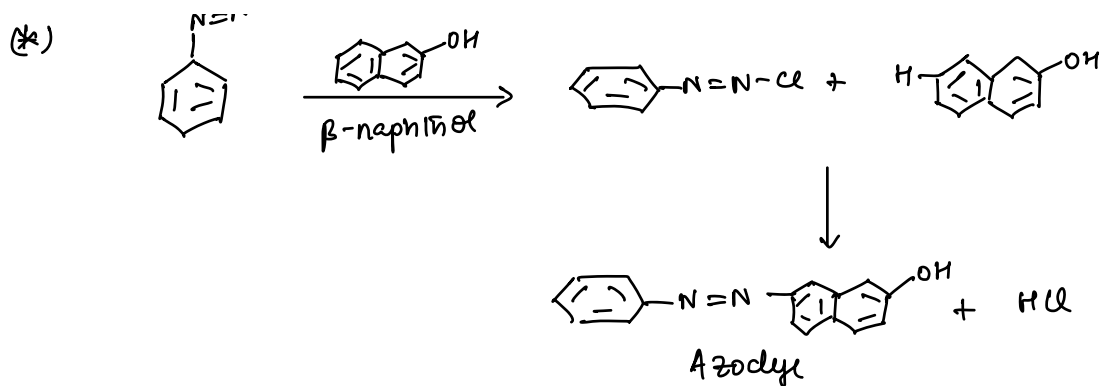


To purify feed stream from sulphur impurities Hydrosulphatization process.
 $(H_2S \uparrow)$

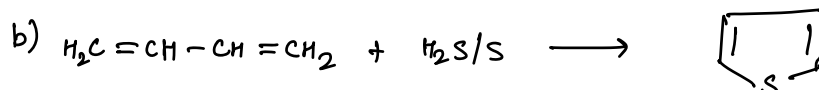
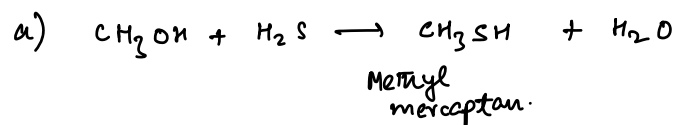
Diazotization rx'n (Paint manufacturing Industries)



...N-Cl

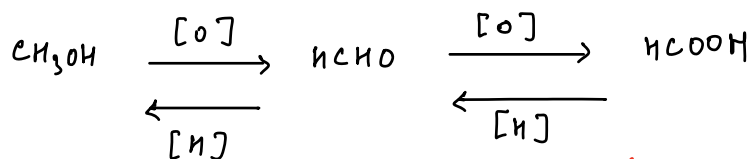
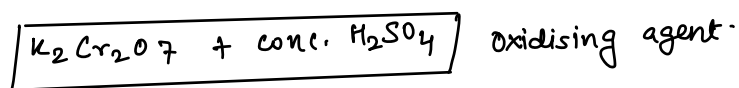


(*) Thionation Rx'n ($-\text{S}, -\text{SH}$)

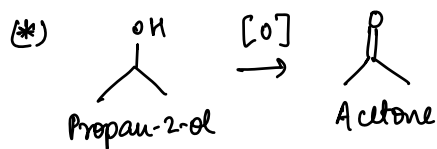


Polymer Industry.

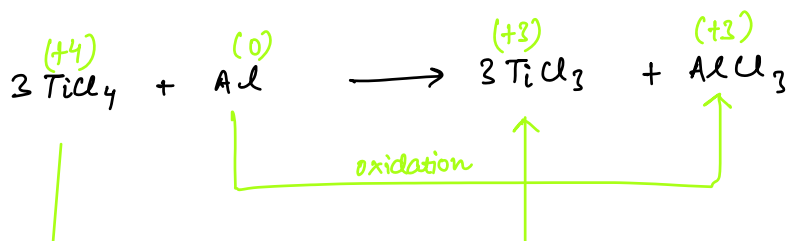
(*) Oxidation Rx'n



LiAlH_4 (cannot use $\text{Sn}/\text{conc. HCl}$ or $\text{Zn}/\text{conc. HCl}$)



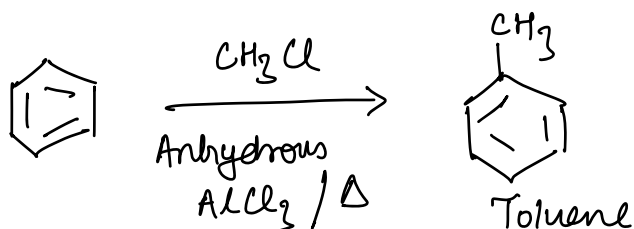
Reduction Rx'n



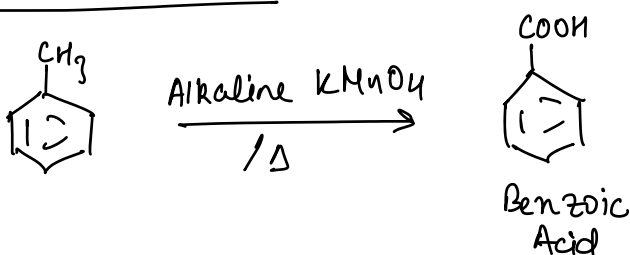
AlCl_3 can be used as a catalyst.

reduction

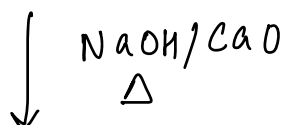
Friedel - Craft's Alkylation Rx'n



Oxidation Rx'n

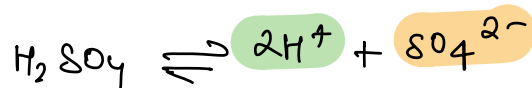
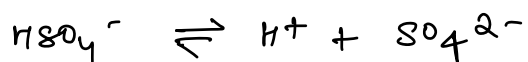
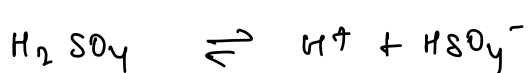
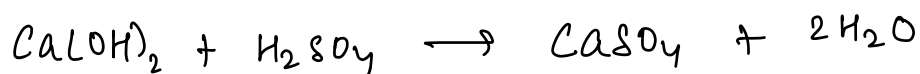


carboxylation rx'n.

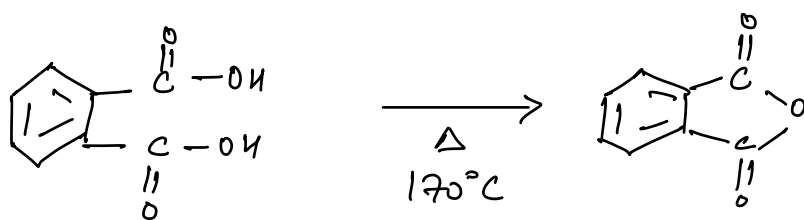


decarboxylation rx'n

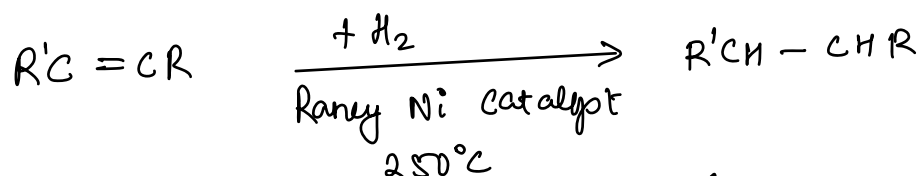
Double Displacement Rx'n (Cement manufacturing industry)



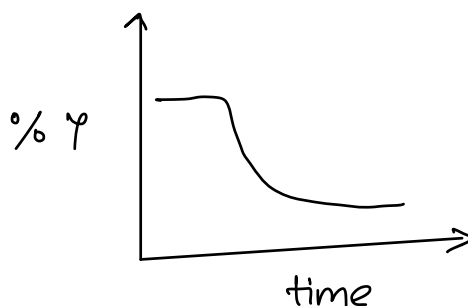
(*) Dehydration Rx'n



(*) Hydrogenation Rx'n

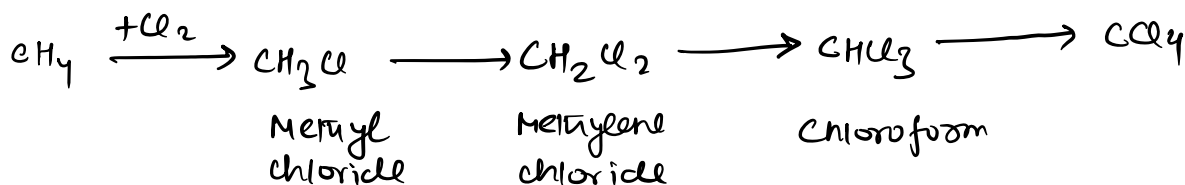
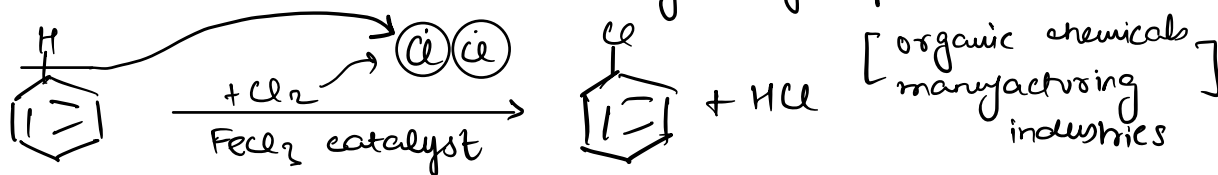


$\text{R}, \text{R}' \equiv \text{H/alkyl functional grp.'s}$

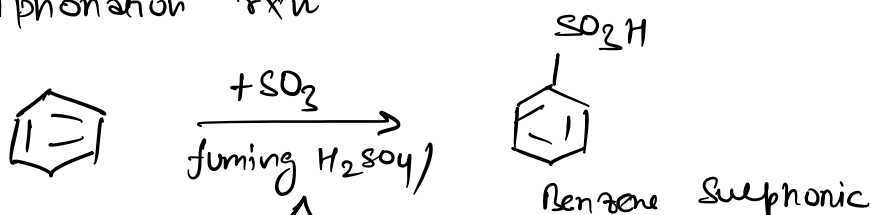


Book:- Dryden's Outlines of Chemical Technology.

a) chlorination rx'n - introduction of -Cl group.

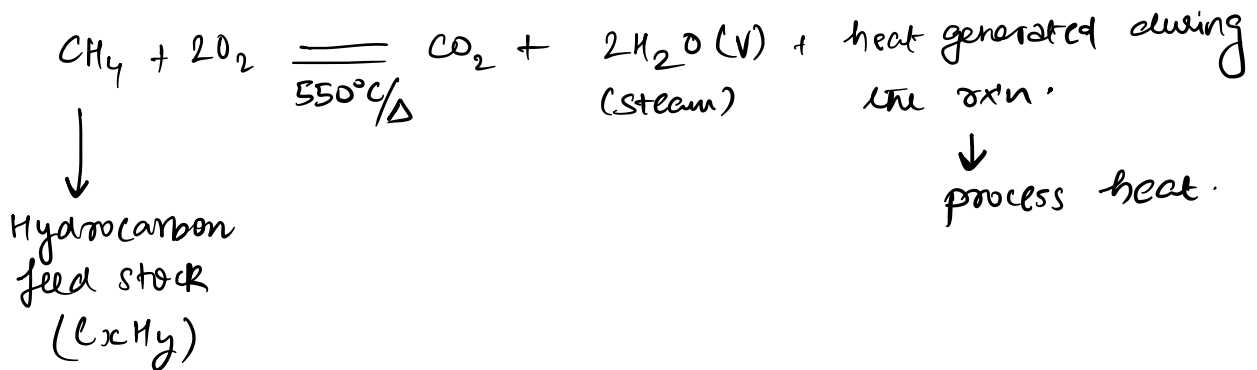


b) sulphonation rx'n



= acid
(sulfonic acid functional groups).

c) combustion rx'n



Catalyst :-

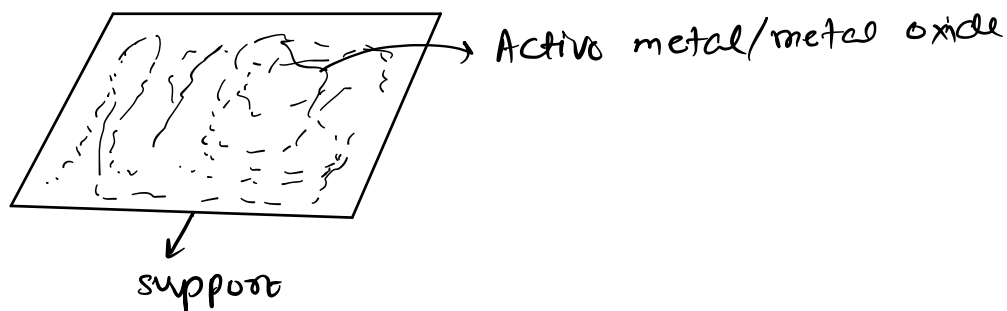
catalyst + it's deactivation

↓ (support + active metal)

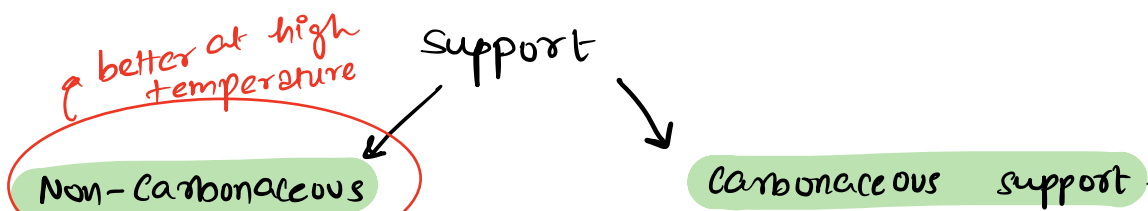
a) supported catalyst

b) unsupported catalyst (Active metal/metal oxides).

(#)



Dispersion - fraction of active metal/metal oxide deposited on the surface of a support.



① SiO_2

② Al_2O_3

③ $\text{SiO}_2\text{-Al}_2\text{O}_3$

④ ZrO_2 [Zirconia]

⑤ Molecular sieve

⑥ Zeolite

① Activated Carbon

② Activated Carbon Fibre (ACFs)

③ carbon nanotubes (CNTs)

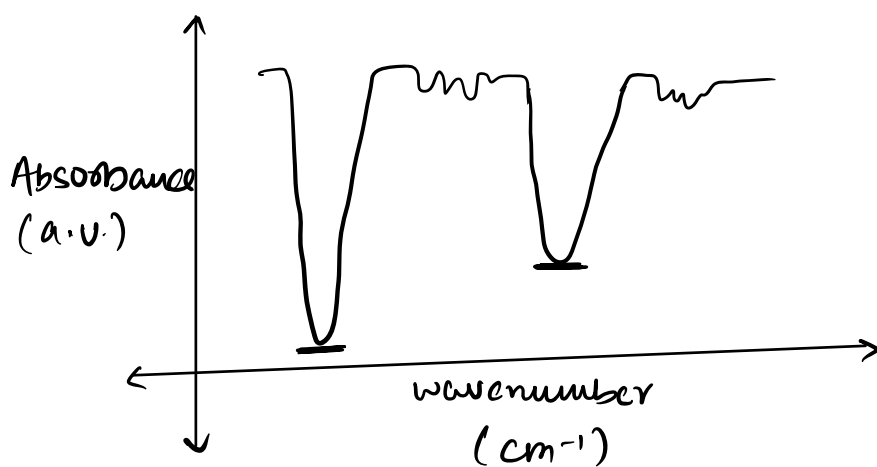
④ Carbon nanofibres (CNFs)

⑤ Graphene (G)

⑥ Graphene Oxide (GO)

⑦ Reduced graphene oxide.
(Less oxygen containing functional groups.)

FTIR → Fourier Transform Infrared Radiation.



(*) At very high temp. burn off takes place ($+ \text{CO}_2 \uparrow$)
that is why we use non-carbonaceous support.

$\text{NiO/AC} \rightarrow \text{NiO}$ (will be left).

support

commercially available :-

a) Merck

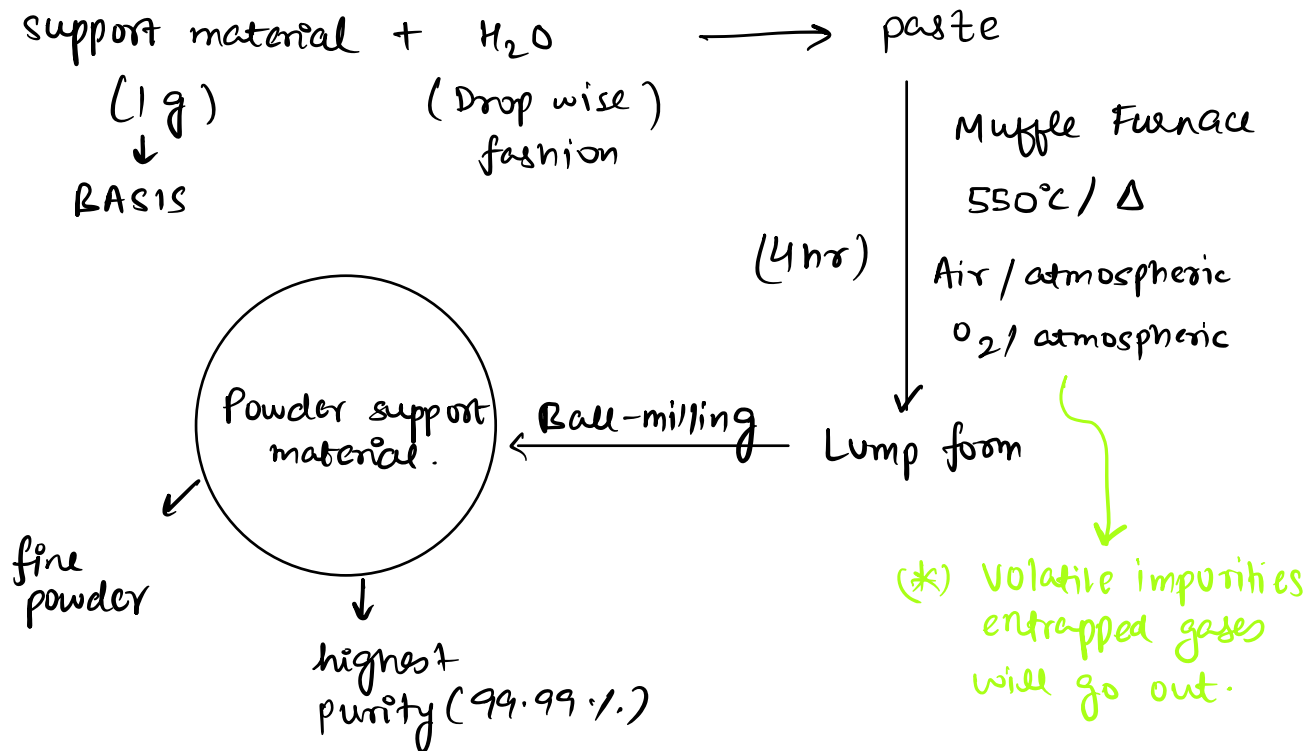
b) Sigma-Aldrich

c) Alfa-Aesar

→ Impure in nature (99.99%)
↑ desired

↓
(90-95%)
↓
purification.

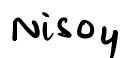
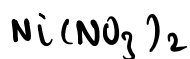
Steps (NON-CARBONACEOUS SUPPORT MATERIAL)



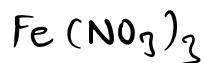
Introduce Active Metal / Metal Oxide

Incipient wetness impregnation method:-

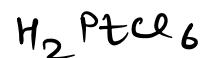
① Precursor salt sol'n



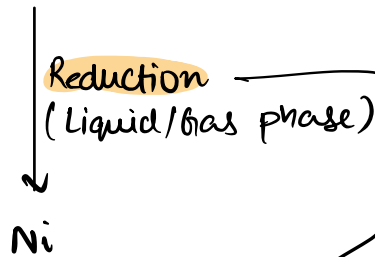
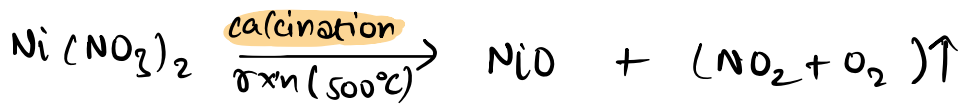
For Ni



For Fe



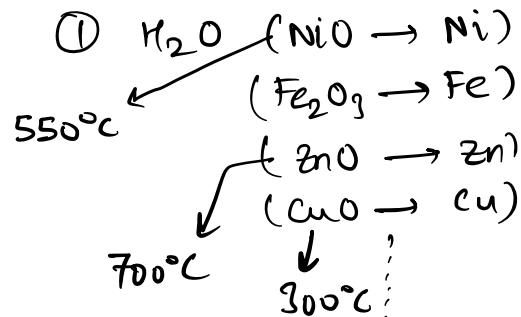
For Pt



Liquid Phase

- a) NaBH_4
- b) Hydrazine mono hydrate
($\text{N}_2\text{H}_4 \cdot \text{H}_2\text{O}$)

Gas phase



Nickel in it's
elemental form

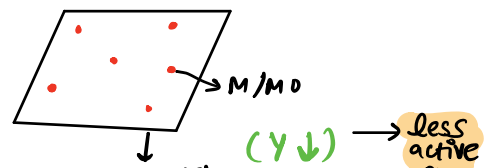
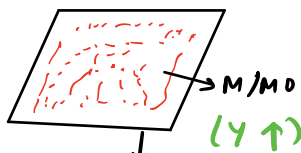
Composite Catalyst

support
↓
purification
of support
material
99.99%
powder form

Active metal / metal oxide (depends on type of application).

AIM: To disperse the active metal/metal oxides onto the support material.
uniform and homogenous.

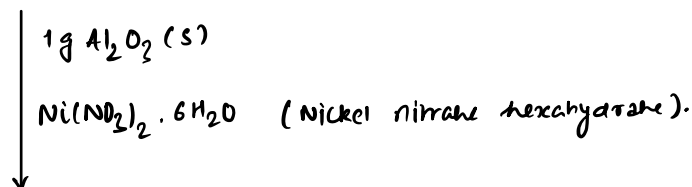
M/MO → ACTIVE
 → NON-ACTIVE



support
(UNIFORM/HOMOGENEOUS)

support sites.
(NON UNIFORM DISPERSION).

BASIS: 1g of support (powder form, 99.99% pure)



① Add $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + \text{H}_2\text{O}$ and mix them thoroughly
 (Aqueous $\text{Ni}(\text{NO}_3)_2$ sol'n).

② Mixing of Al_2O_3 and aqueous $\text{Ni}(\text{NO}_3)_2$
 ↓ dropwise manner; thorough mixing.

$[\text{Ni}(\text{NO}_3)_2 / \text{Al}_2\text{O}_3]$
 paste.

muffle furnace
 $500^\circ\text{C}, 4\text{h}, \text{O}_2$
 (CALCINATION).

$\text{NiO} / \text{Al}_2\text{O}_3$

H_2 -Reduction
 $550^\circ\text{C}, 2\text{h}$

INCONEL
 (High grade stainless
 steel)
 (leak free).

$\text{Ni} / \text{Al}_2\text{O}_3$

③ composite catalyst

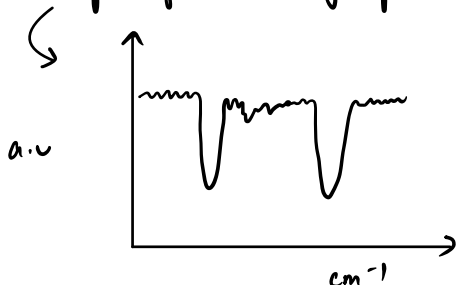
↓ experiments

physicochemical properties

a) surface area and pore size distribution. (BET method)

b) Surface acidic/basic properties. (TPD Analysis) → Temperature programmed desorption analysis

c) Surface functional groups (FTIR)



(450 to 4000 cm^{-1}) [Already discussed previously].

Surface area and PSD

Pore Diameter

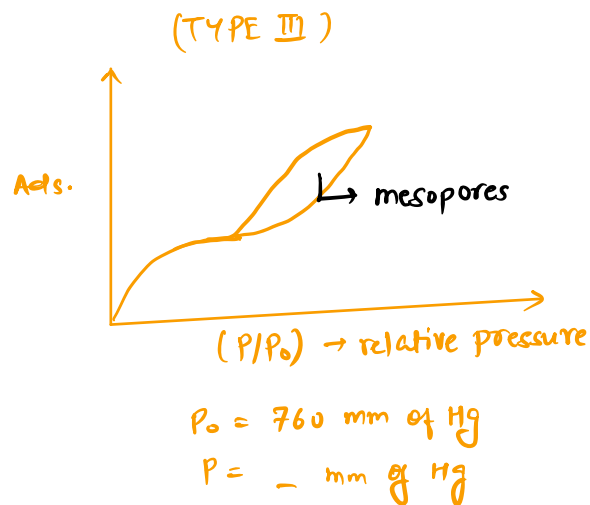
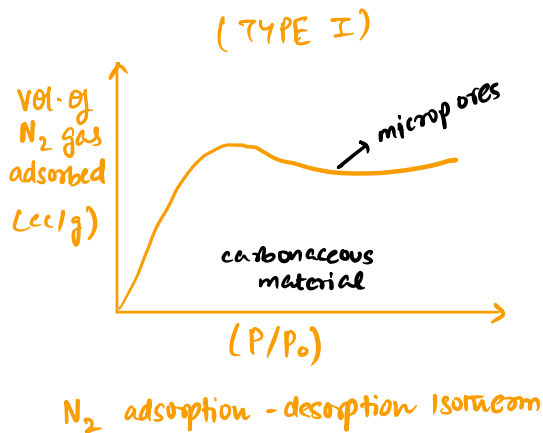
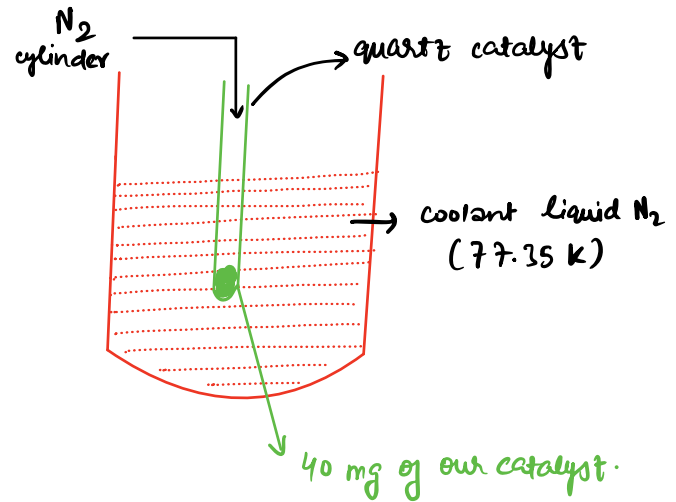
3 different types of pores

→ micropores < 2 nm
 → mesopores 2-50 nm
 → macropores > 50 nm

surface area is measured in m^2/g

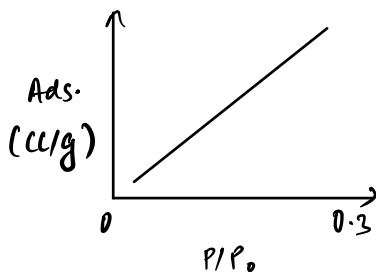
BET

- ① Surface area (m^2/g)
- ② Total pore volume (cc/g)
 - ↓
 - Σ (micro + meso + macro) pores
 $\text{cc/g} \quad \text{cc/g} \quad \text{cc/g}$
- ③ Avg. pore diameter ($\text{\AA}/\text{nm}$)



For finding macropore volume

$$\Rightarrow \text{Macropore volume} = \text{Total pore volume} - [\text{Micropore volume} + \text{Mesopore volume}]$$



(In BET, we take till 0.3 because profile is linear here in calculating intercept/slope).

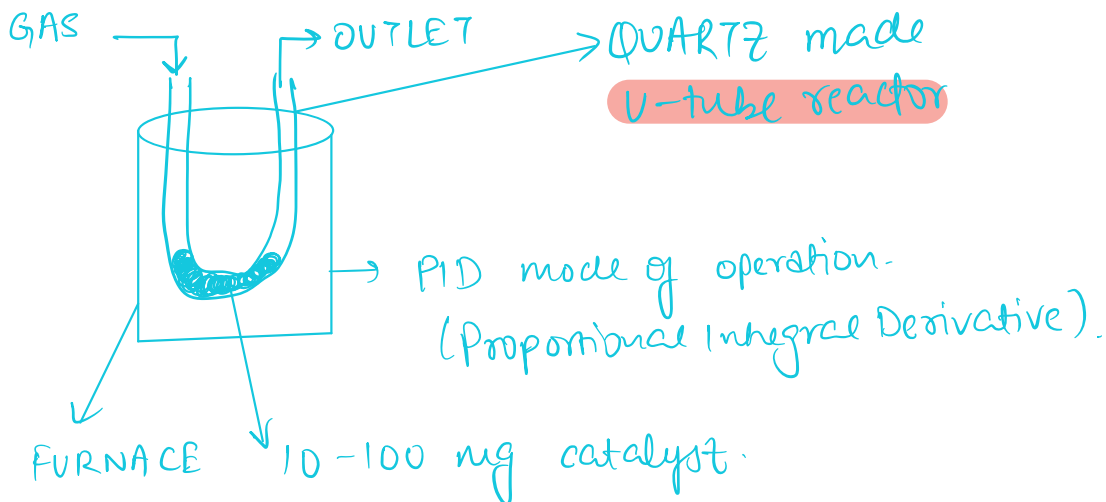
Physiochemical properties of a composite catalyst:-

Different surface functional groups



Surface properties - ACIDIC/BASIC

Temperature programmed desorption (TPD) analysis



NH_3 as a TPD probe gas (to determine acidic sites)

CO_2 as a TPD probe gas (to determine basic sites)

NH_3 will get absorbed

ACIDIC

CO_2

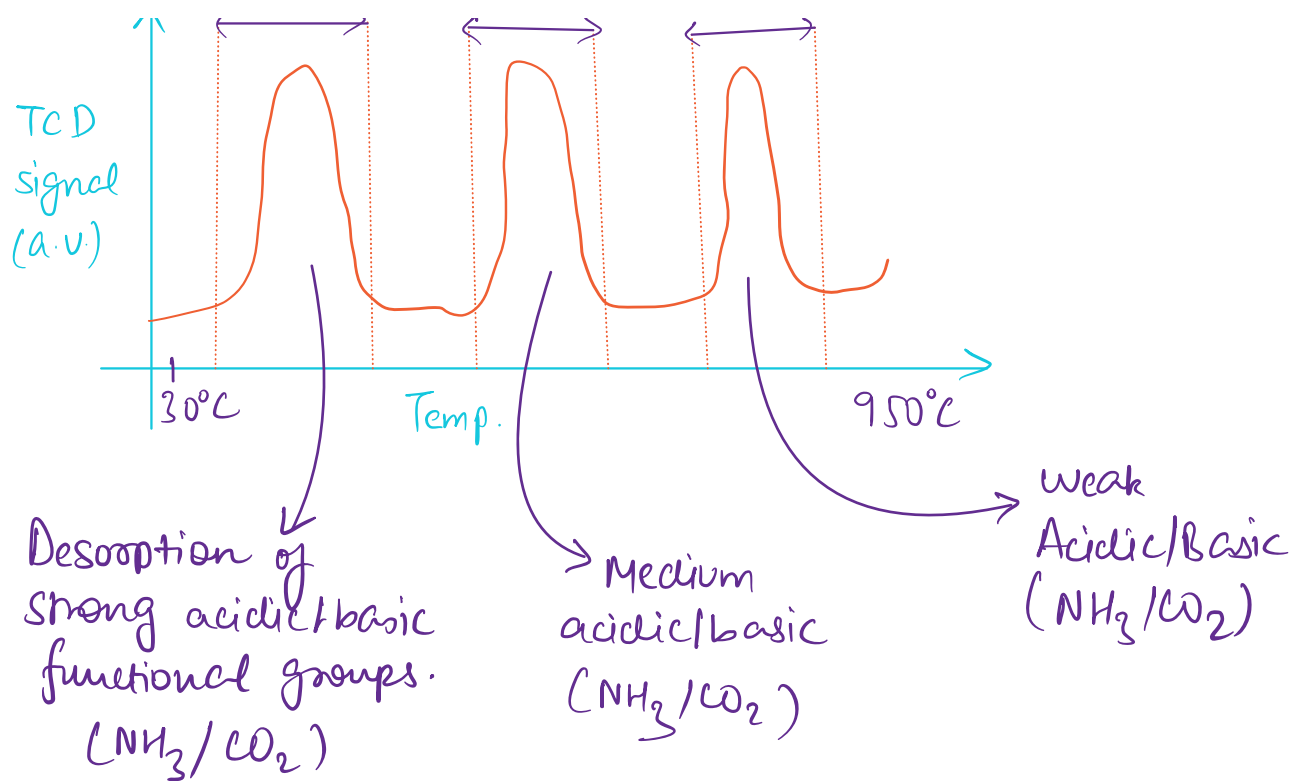
BASIC

Outlet of U-tube $\rightarrow$ TCD (m/z ratio).
(Thermal Conductivity)
Detection

Furnace conditions:-

$\rightarrow 30^\circ\text{C}$ to $950^\circ\text{C}/1000^\circ\text{C}$ @ $10^\circ\text{C}/\text{min}$.

Less temp. zone Intermediate temp. zone High temp. zone



Catalyst Deactivation - when catalyst lose their activity.

↓

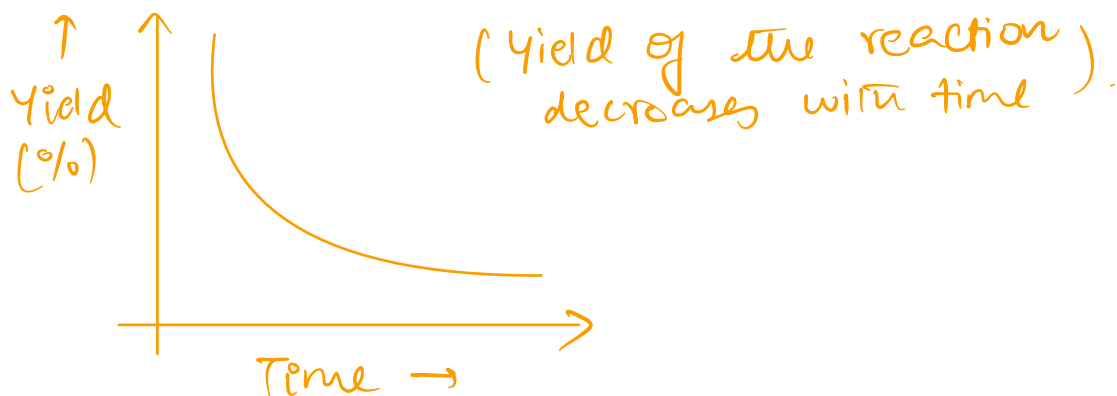
COOKING

↓

SINTERING

↓

POISONING

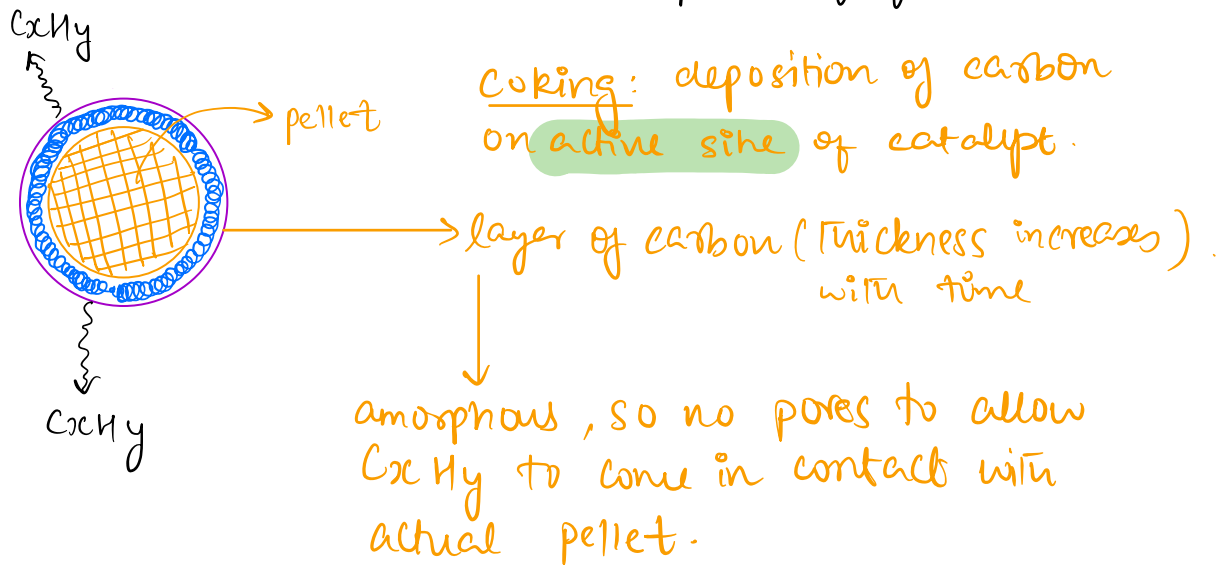
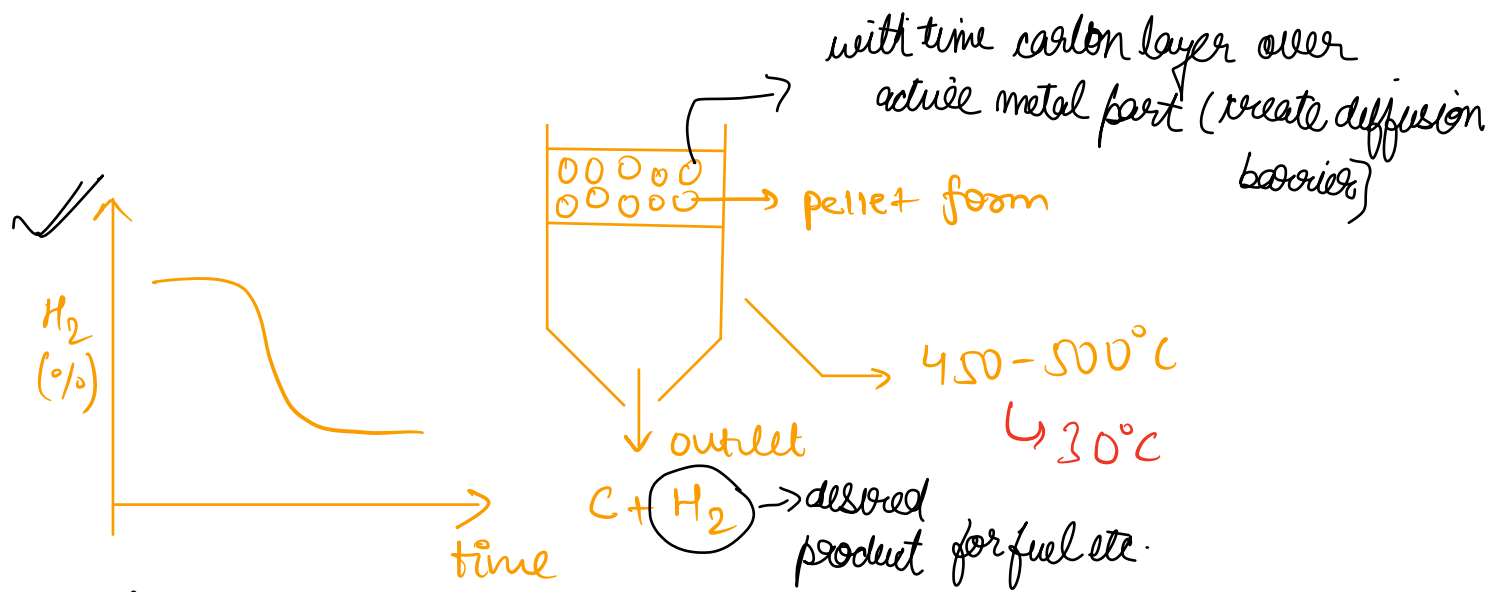


(a) COOKING :



→ Packed bed reactor

↳ deposition of amorphous carbon on active sites of a catalyst



So, no reaction occurs and yield of H_2 decreases-
 $CxHy$ comes out as it is.

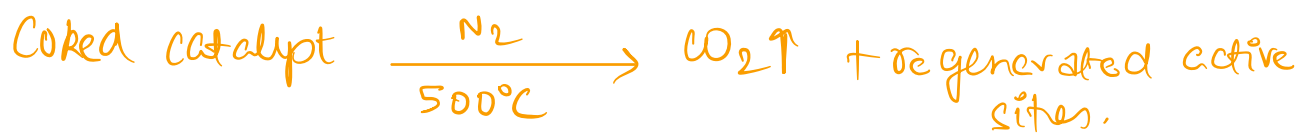
Regeneration of catalyst:

by reactivating the surface sites

In O_2/N_2 environment.

Steam activation.





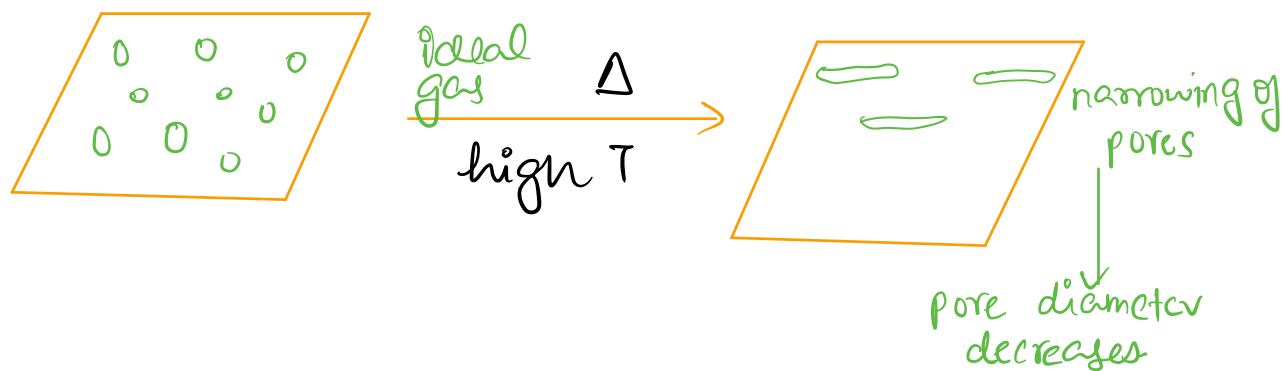
In steam activation, $\text{H}_2\text{O}(\text{steam}) + \text{N}_2 \xrightarrow{900^{\circ}\text{C}}$
 Again CO_2 is produced. $\downarrow$
 carrier gas

TGA analysis - N_2 environment, $30^{\circ}\text{C} \rightarrow 1150^{\circ}\text{C}$
 [Thermogravimetry]

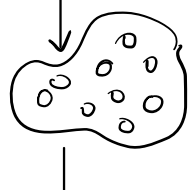
(b) SINTERING: At high temperatures, $> \sim 600^{\circ}\text{C}$.

$\swarrow$
 narrowing of pores.

$\searrow$
 aggregation of active metals.



REACTANT



(bulk phase)

POROUS
CATALYST

Steps of catalytic reaction:-

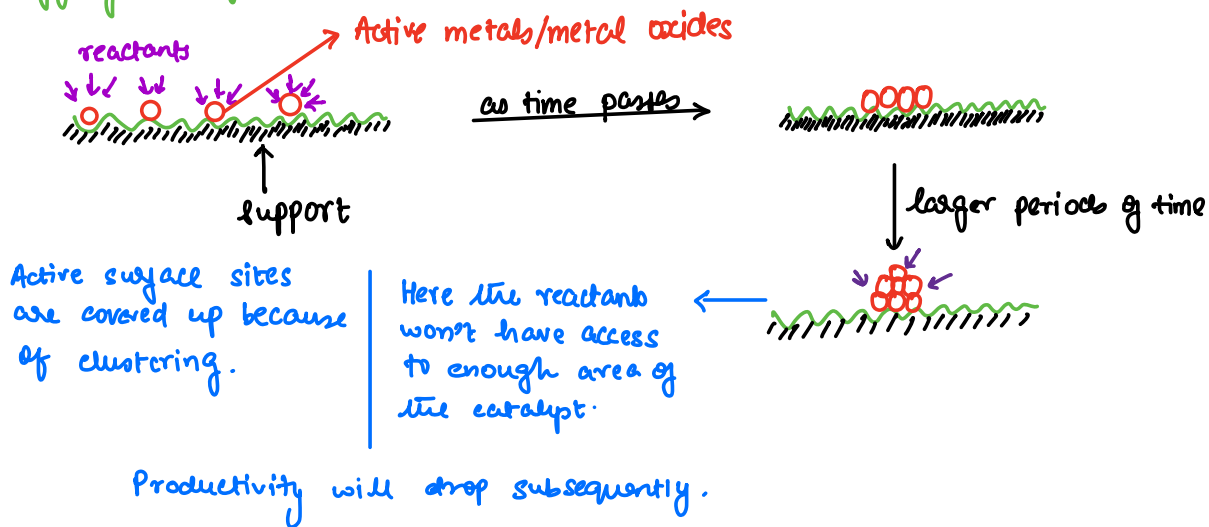
- ① Diffusion of R from bulk to external surface.
- ② From external surface to pore mouth.

↓
PRODUCT

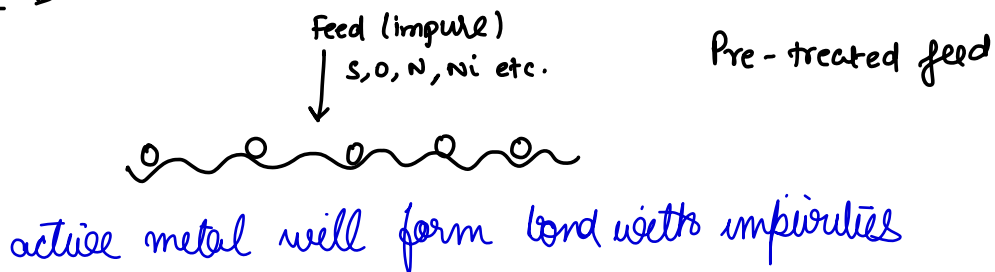
- ③ From pore mouth to inner surface of pore.
- ④ Reaction
- ⑤ Products diffuse from inner pore surface to pore mouth.
- ⑥ Pore mouth to external surface.
- ⑦ External surface to bulk phase.

0.1 to 0.4 times lesser than the melting point of the catalyst
→ to avoid narrowing of pores in the catalyst, and also to avoid aggregation.

Aggregation of solids:



Poisoning ⇐



Sulphur manufacturing:

→ S is the basic raw material for H_2S/H_2SO_4 production.

→ properties of sulphur:

Atomic weight = 32.07

→ Allotropic forms $\begin{cases} \rightarrow \text{Rhombic (112.8}^\circ\text{C MP)} \\ \rightarrow \text{Monoclinic (119}^\circ\text{C MP)} \end{cases}$

→ Boiling point = 444.6°C

→ Solubility - insoluble in H_2O , soluble in organic solvents and liq. NH_3/NH_4OH .

→ Different forms - Rock, Molten, Lump, Fine Powder.

(#) Consumption Pattern

① SO_2 ② SO_3 ③ H_2SO_4 ④ $H_2S_2O_7$

Filtration is done to remove carbonaceous impurities or acidic impurities.

75-95% of S is consumed in oxide form).

(#) End uses

① SO_2 , SO_3 , H_2SO_4 (-90%), CS_2 , P_2S_5 .

② Gun Powder ($C + S + KNO_3$)

③ Paints and putties containing S.

(#) End uses of H_2SO_4

① fertilizer industries.

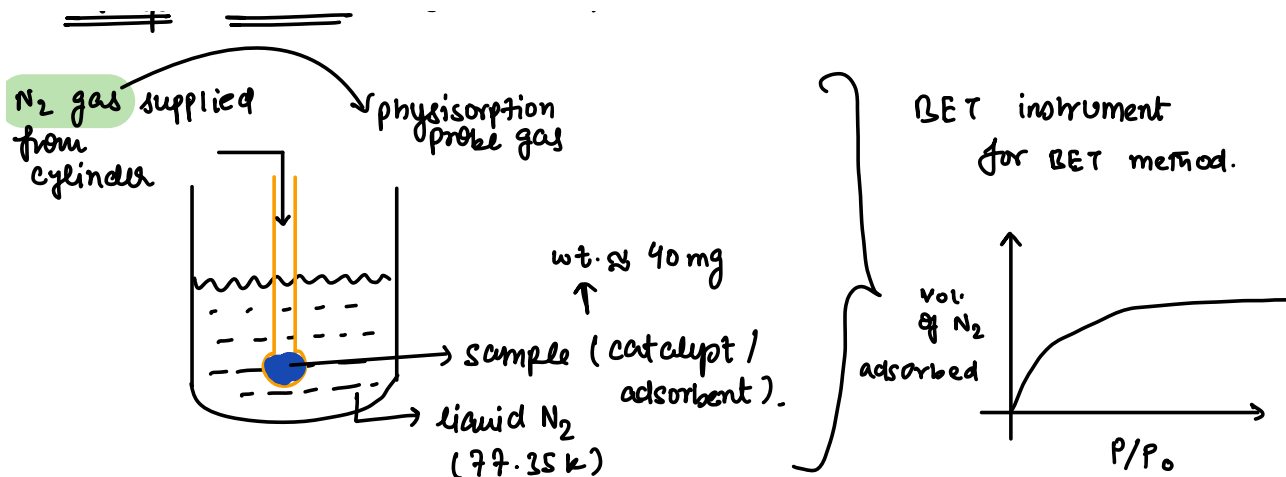
② Inorganic chemical manufacturing industries.

③ Organic chemical manufacturing industries.

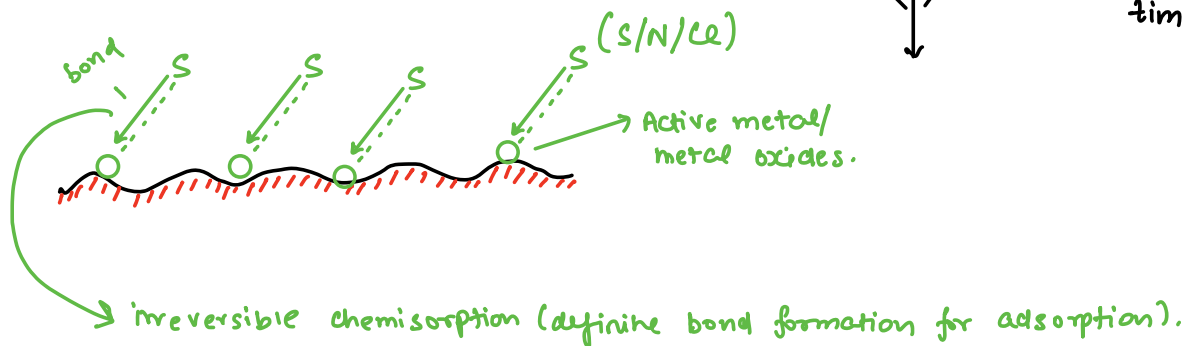
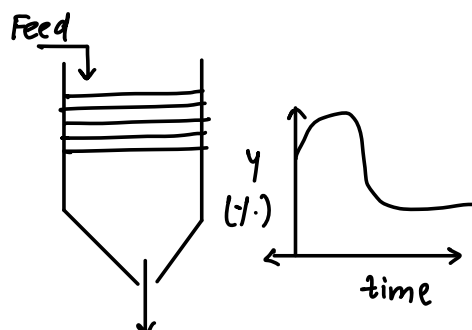
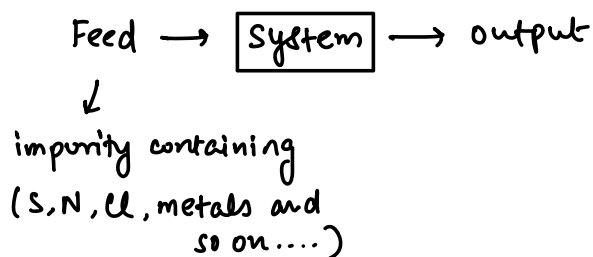
④ Paper and pulp industries.

⑤ Polymer industries.

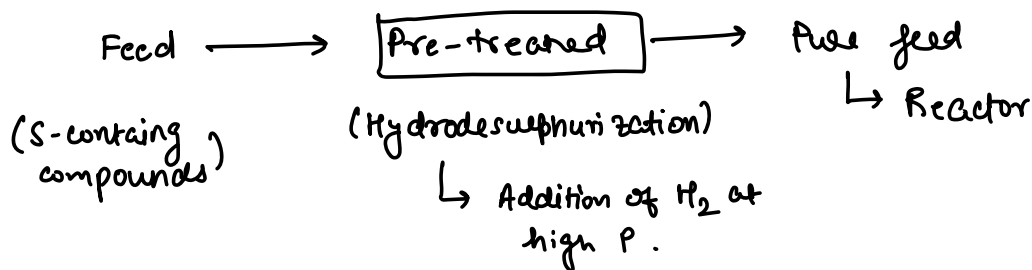
Catalyst Deactivation (Conti.....)

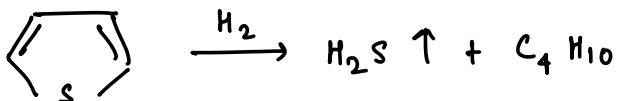


- ① Coking ② Sintering ③ Poisoning

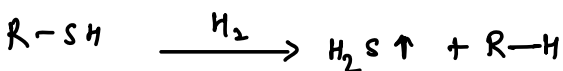


Purification of feed





Thiophene



Mercaptan

Hydrodechlorination



Suphus manufacturing (continued.....)

Synthesis -

- ① Frasch Process (USA)
- ② klaus Process (USA + India)
- ③ Finnish Process (India)

Frasch:

Raw materials - (a) S-bearing sanddome.
(b) Hot water.

Waste water/ effluent obtained from industries. It is contaminated.

If it is paint/dye manufacturing industry, then will paints/dyes in the effluents.

Effluent is totally impure.

(Dyes/paints/pigments/s/ heavy metals like Pb, As, Hg etc.).

```

graph TD
    A[wahe water] -- purification --> B[purified water]
    B -- "furnace/heater" --> C[Hot water.]
  
```

Purification Steps:-

a) Chlorination reaction.

b) Fenton reaction.

c) Adsorption

d) Advanced oxidation process (High P + O₂/Air)
[AOP]

catalytic oxidation process.

⊕ S-sulphides + Hot water → Molten S.

In abundance in US,
but not in India.

Klaus: (oxidation and reduction of H₂S).

Raw Material - H₂S (from natural gas/petroleum refineries).

Reactions -



Step I:

Reactor I → 300-400°C, Bauxite catalyst.
70% - 80% S-conversion.

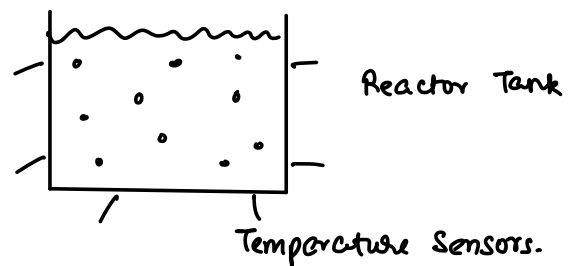
→ Alumina (Al₂O₃)

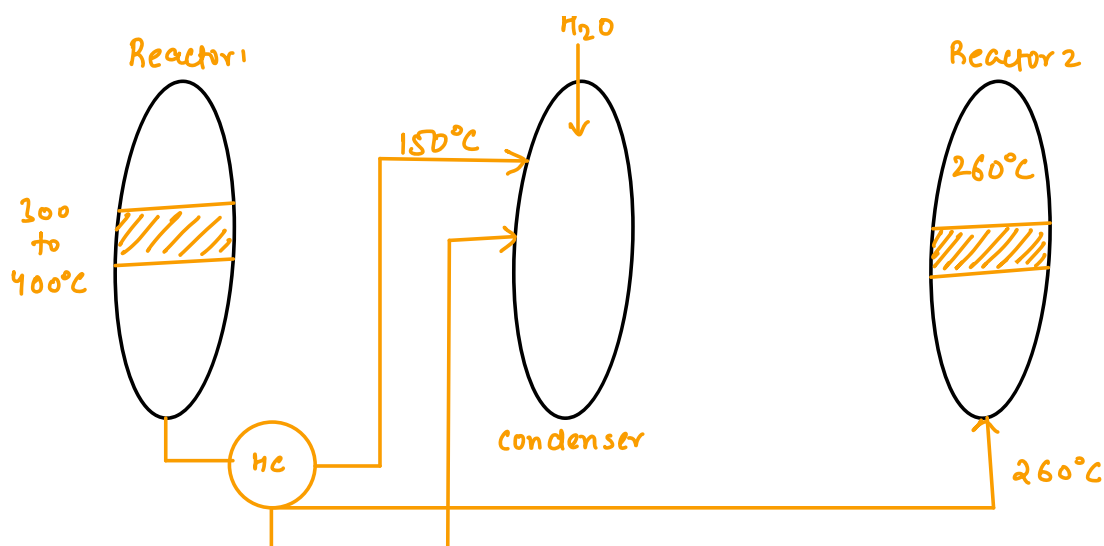
Step II:

Reactor II → 250-300°C, Al₂O₃, 95-99% conversion.

Hotspot in a reactor ⇒ uneven temperature distribution.

To avoid hotspot formation
we use 2 reactors

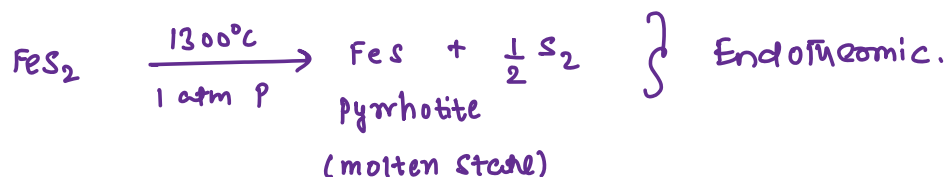




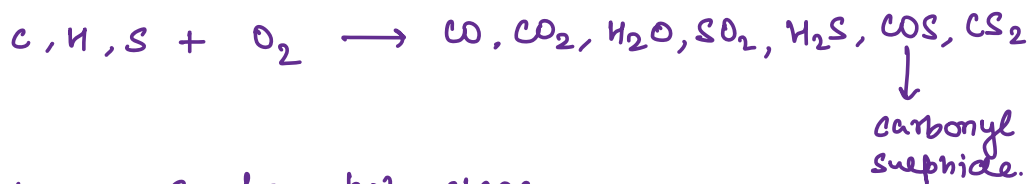
Finnish:

Raw Material - FeS₂ (iron pyrites)
(40-45% S).

① Thermal dissociation of FeS₂.



② General Combustion:



③ Recovery of S from hot-stage
(600°C, Bauxite (Al₂O₃) catalyst).



④ Recovery of S from cold-stage (Klaus Process).



⑤ Roasting of FeS



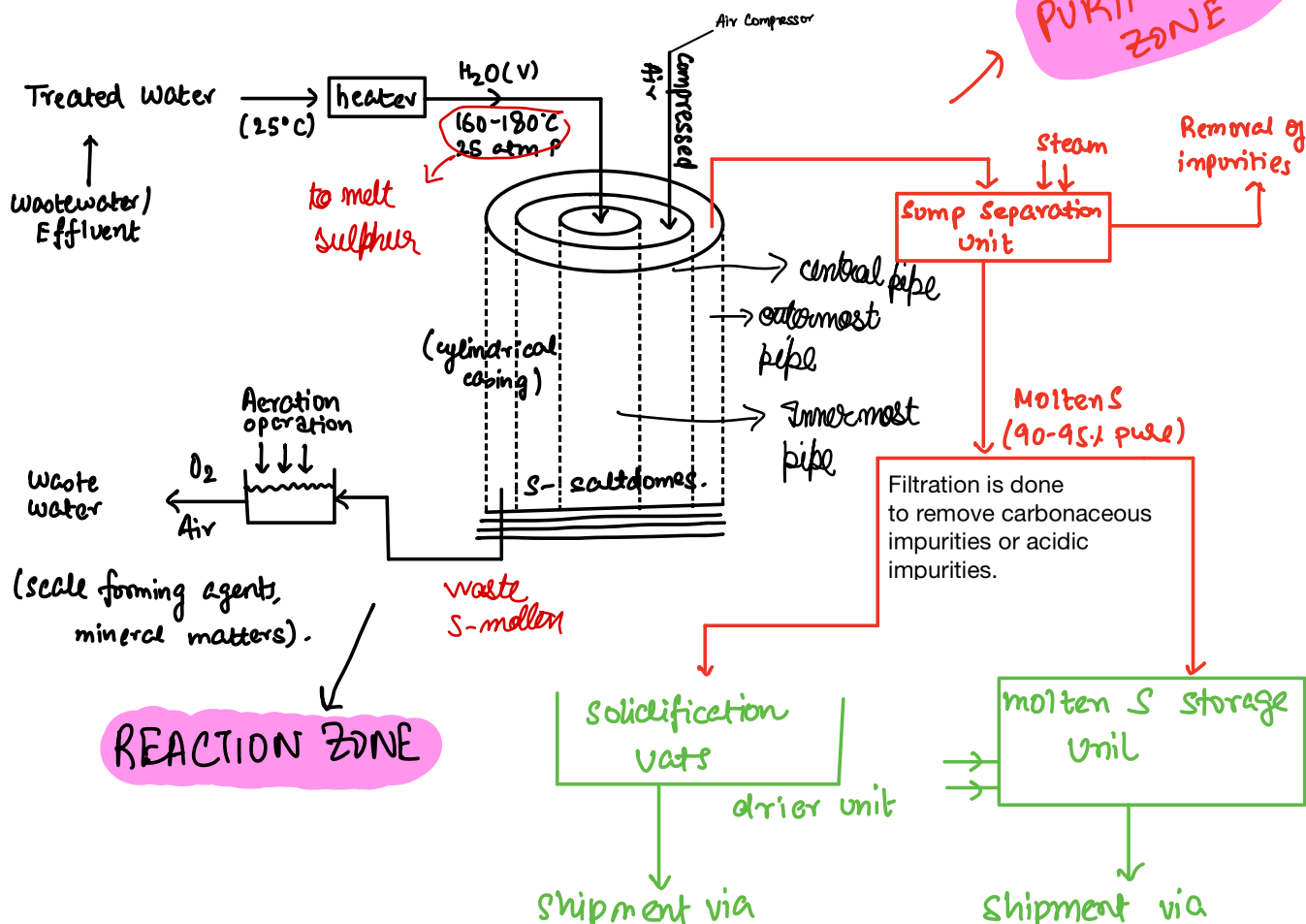
molten state

used in steel manufacturing industries.

$\text{HCHO} \rightarrow$ used in resins

Methanol is also used in producing anti-knock agents.

PROCESS FLOW DIAGRAM (FRASCH PROCESS)



oil/water

inulated buages
and cars.

↓
**TRANSPORTATION
ZONE**

wastewater pre-treated by:

- Cl_2
- Adsorption
- Fenton Reaction
- Advanced oxidation

Major engineering problems:

- Corrosion
- Suitable sources of waste water.
- Heat-loss during transport of molten-S.

PROCESS FLOW DIAGRAM (KLAUS PROCESS)

